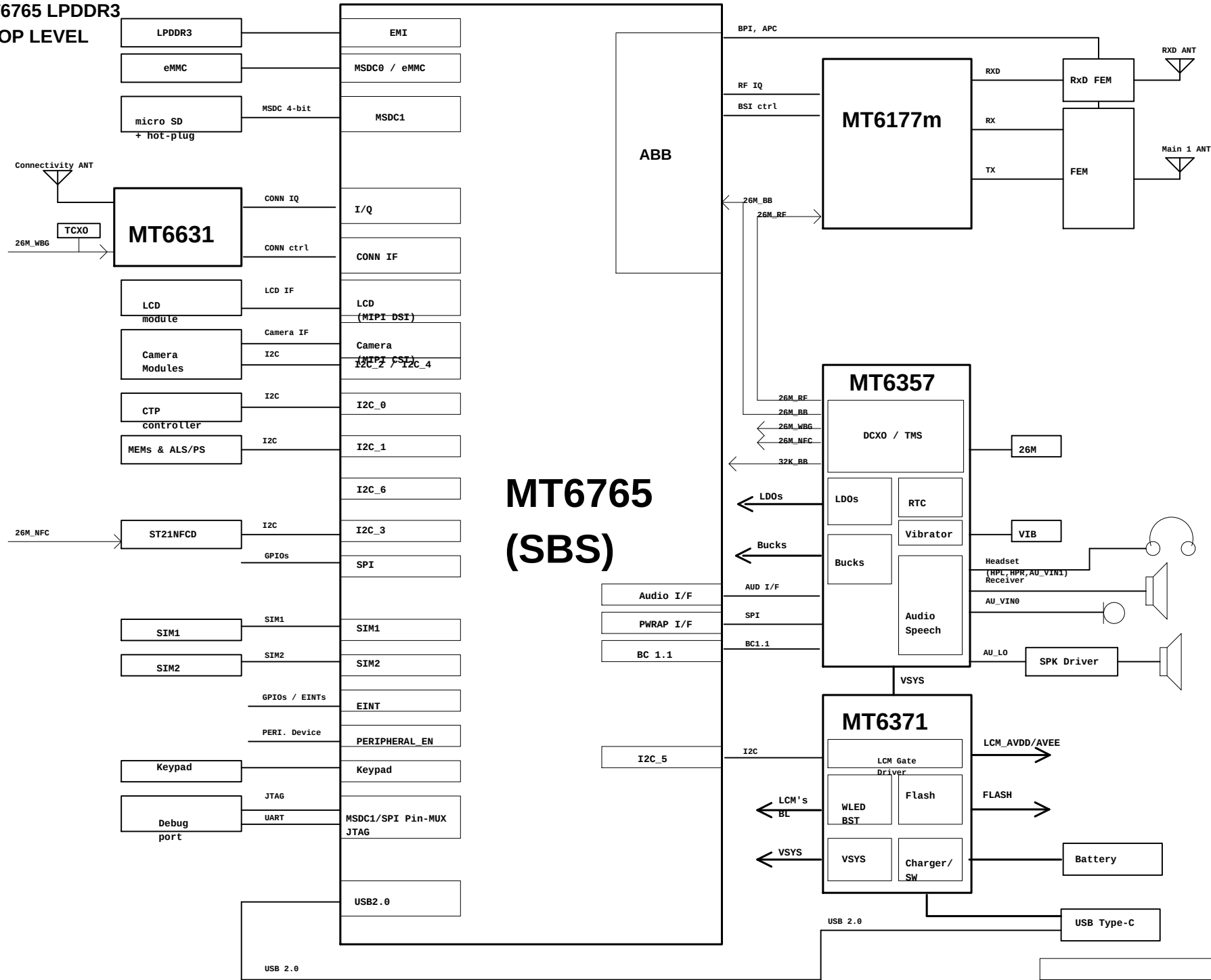


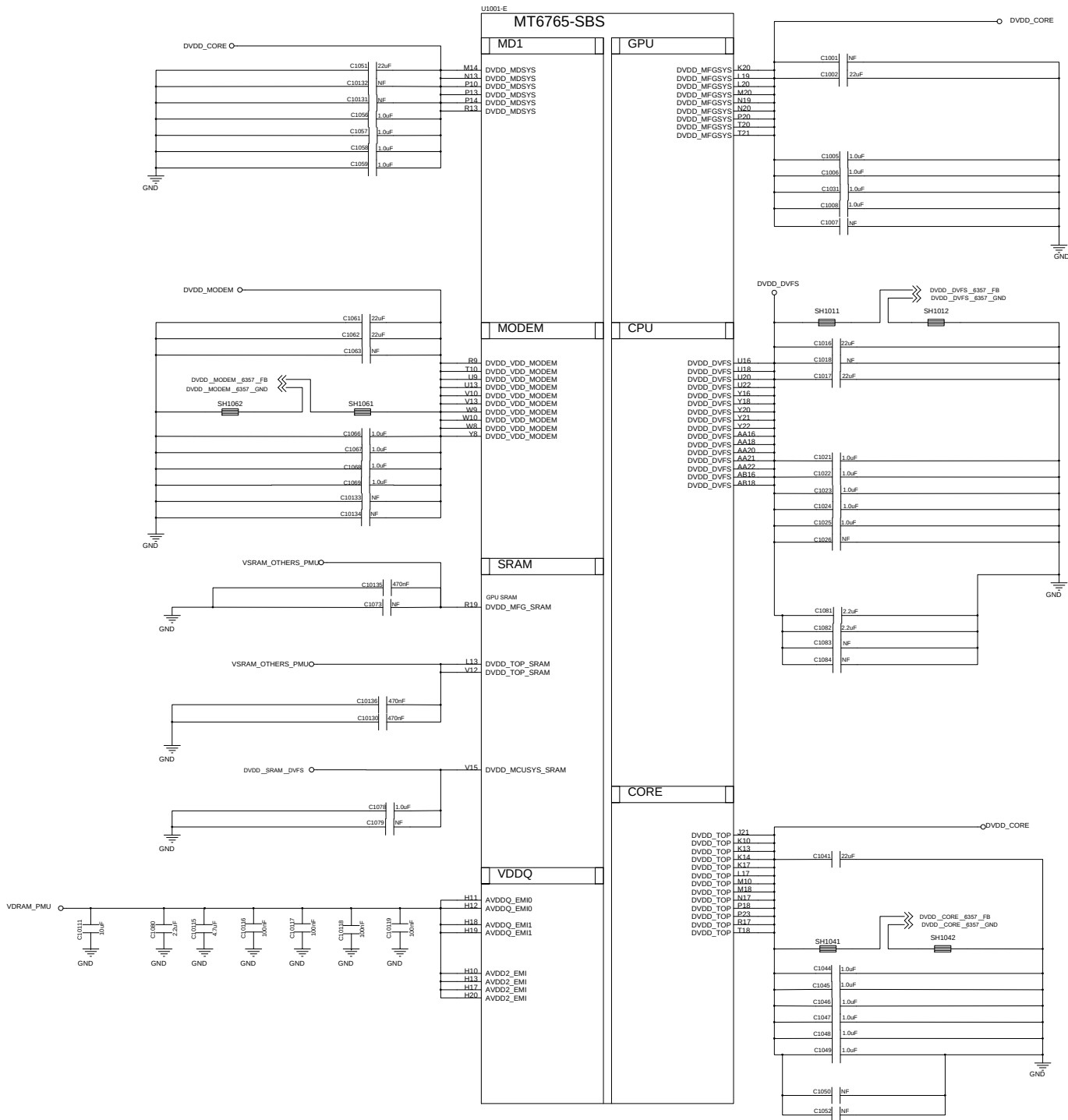
Project : MT6765 LPDDR3
REF_SCH TOP LEVEL

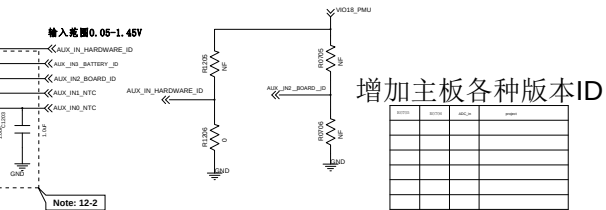
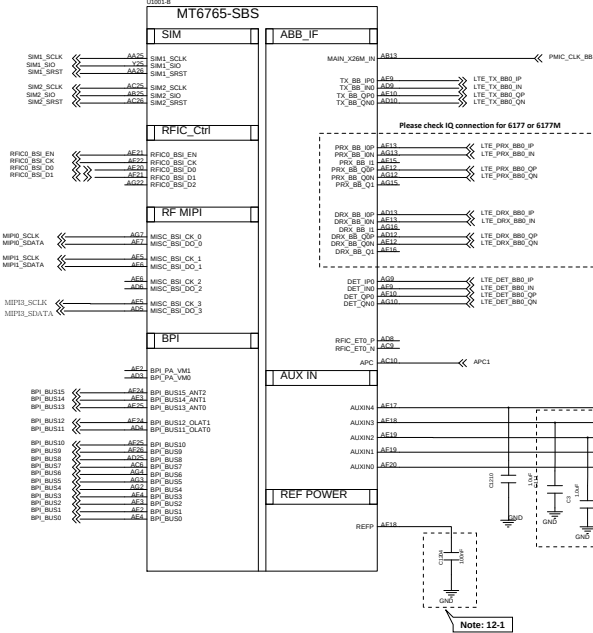
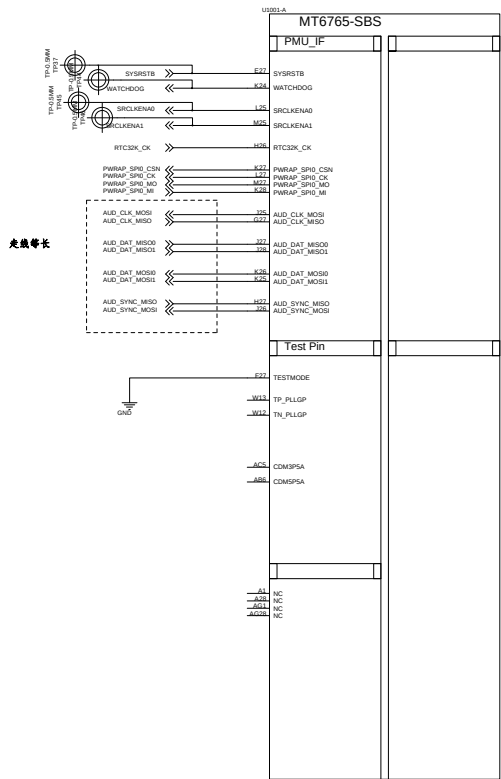


I2C	Sub SYS	Function	Part Number	I2C Spec.		i2C Slave Address / Write / Read (7-bit mode)	
I2C-0	AP	Cap Touch controller	GT1151	400 Kbps		0x5D	Write:0xBA / Read:0xBB
I2C-1 (I3C)	AP Sensor Hub	Magnetic Sensor	AK09918C	400 Kbps		0x0C	Write:0x18 / Read:0x19
		Ambient Light Sensor	CM36558	400 Kbps		0x51	Write:0xA2 / Read:0xA3
		Proximity Sensor					
		Pressure Sensor	BMP280	400 Kbps		0x77	Write:0xEE / Read:0xEF
I2C-2 (I3C)	AP	Rear Camera	IMX230	400 Kbps		0x1A	Write:0x34 / Read:0x35
			EEPROM	400 Kbps		0x50	Write:0xA0 / Read:0xA1
			AF driver	400 Kbps		0x0C	Write:0x18 / Read:0x19
I2C-3	AP	Audio Smart PA	RT5510	400 Kbps		0x34	Write:0x68 / Read:0x69
		NFC	ST21NFCD	400 Kbps		0x08	Write:0x10 / Read:0x11
I2C-4 (I3C)	AP	Front Camera	S5K2T7	400 Kbps		0x2D	Write:0x5A / Read:0x5B
			EEPROM	400 Kbps		0x52	Write:0xA4 / Read:0xA5
			AF driver=NA				
I2C-5	AP	Sub-PMIC	MT6371 PMU	3.4 Mbps		0x34	Write:0x68 / Read:0x69
			MT6371 PD	3.4 Mbps		0x4E	Write:0x9C / Read:0x9D
I2C-6	AP						

Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

Date	Category	Item
2017.11.24 (V0.1)	Page 05	V0.1 Release
2017.12.7 (V0.2)	Page 11	Change power of AVDD18_DDR(H21) from VI018_PMU to EMI_VDD1, connecting EMI_VDD1 to SH2102 in star connection
	Page 12	Add Note 12-5
	Page 21	Add SH2102 for star connection among EMI_VDD1, AVDD18_SOC, and VI018_PMU
	Page 22	
	Page 44	1. Change C2304 from C / 1 / uF / 10V to C / 1 / uF / 6.3V 2. Change R2301 from R / 1.5 / K to R / 7.5 / K 3. Update the L22D53 power off sequence, change C4422 from 0.1uF to 2.2uF, and change C4423 from 0.1uF to 1uF
	Page 51	2. Change VDD1 power of eMCP from VI018_PMU to EMI_VDD1, fulfilling power rail in star connection 1. Add 2nd source plan for U5004 2. Add Note 51-3 and Note 51-4





Schematic design notice of "12_BB_1" page.

Note 12-1: The de-coupling cap. for REFP (AF18 ball) have to be placed as close to BB as possible.

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: "PWRAP_SPIO_CSN" and "AUD_DAT_MOSIO" are bootstrap pin to select which interface will be the JTAG pin out.

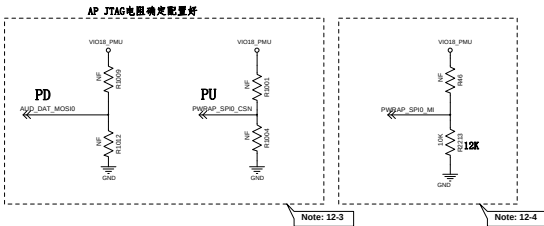
PWRAP_SPIO_CSN	AUD_DAT_MOSIO	JTAG Function	
default=PU	default=PD	AP_JTAG	MD_JTAG
HI	LO	N/A	N/A
HI	HI (by ext. PU)	SPIO+EINT8	SP1+SP3
LO (by ext. PD)	LO	SPIO+EINT8	N/A
LO (by ext. PD)	HI (by ext. PU)	MSDC1	N/A

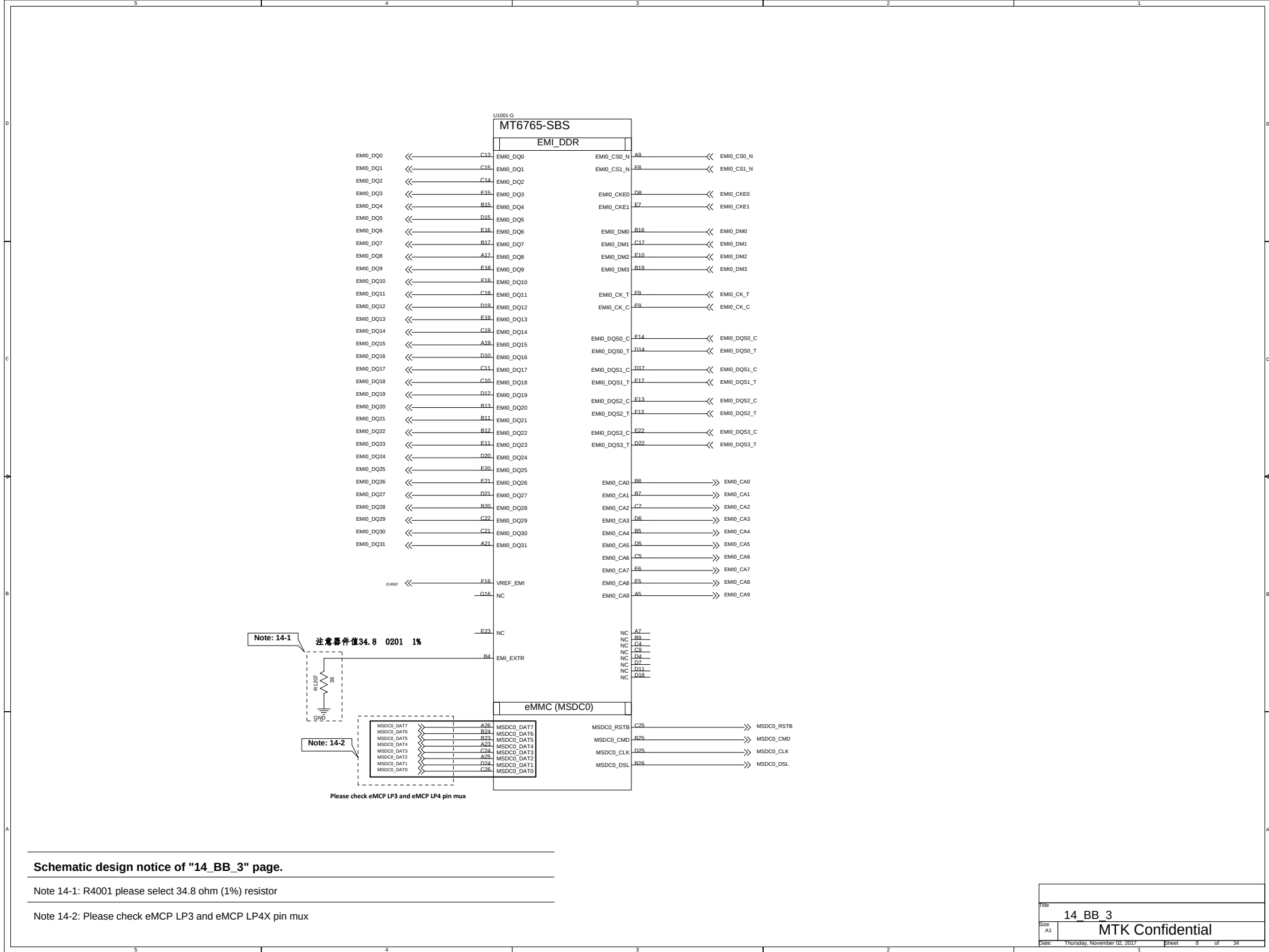
Note 12-4: PWRAP_SPIO_MO and PWRAP_SPIO_MI are DDR type feature in bootstrap

PWRAP_SPIO_MI	Booting interface
default=PU	DDR
LO (by ext. PD)	LPDDR3 follow LP3 Ref SCH.
HI	LPDDR4X follow LP4X Ref SCH.

注释错误

Note 12-5: Please set unused IQ pins in NC

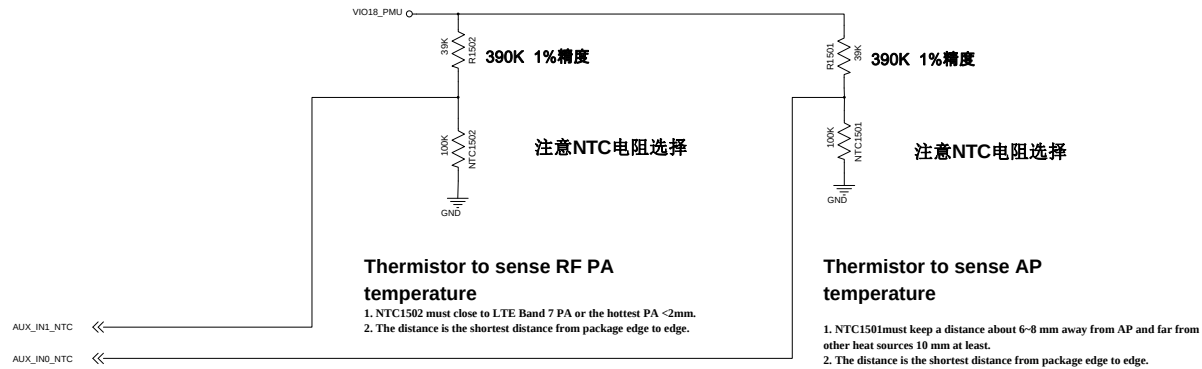


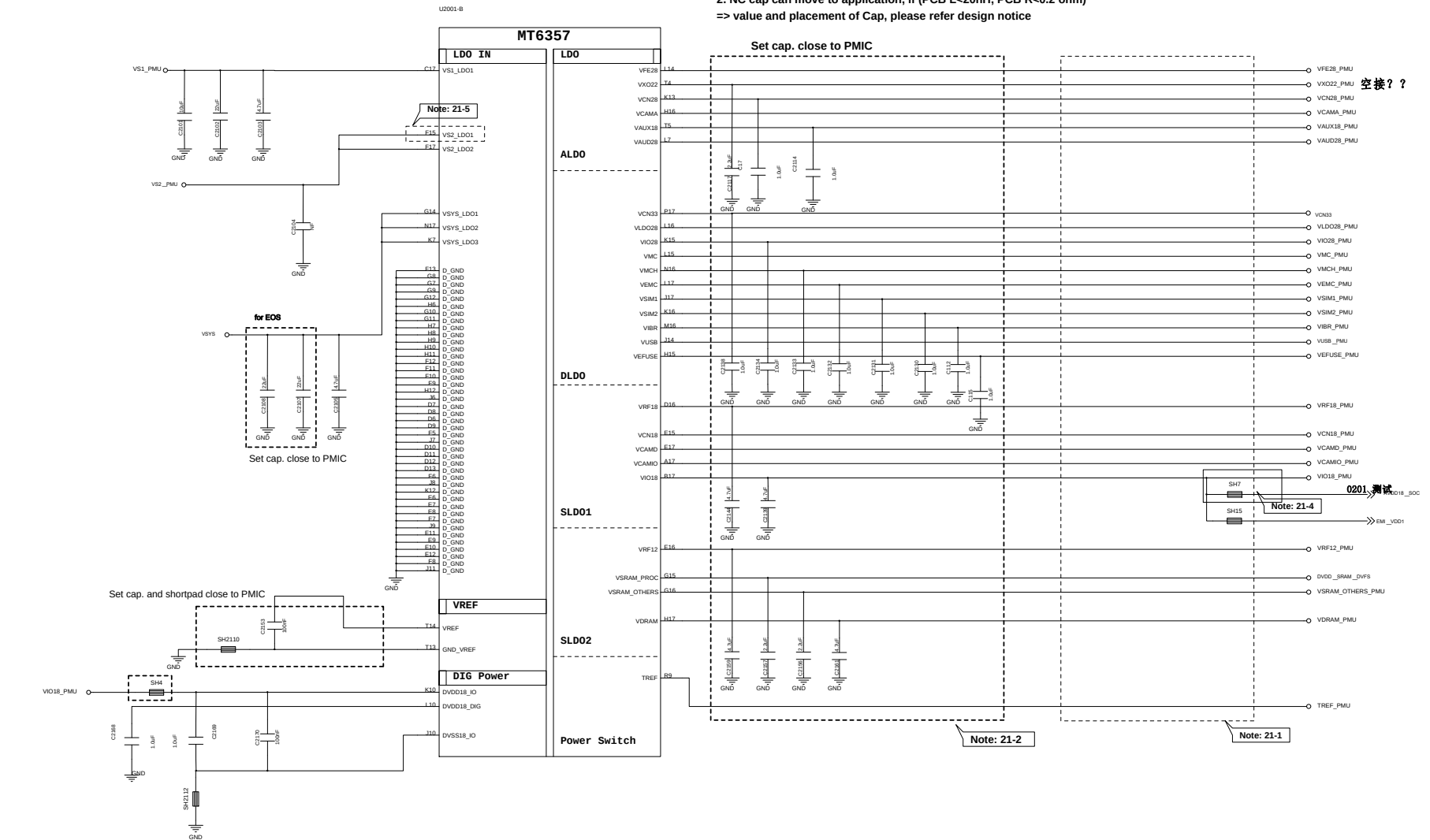


Schematic design notice of "14_BB_3" page.

Note 14-1: R4001 please select 34.8 ohm (1%) resistor

Note 14-2: Please check eMCP LP3 and eMCP LP4X pin mux



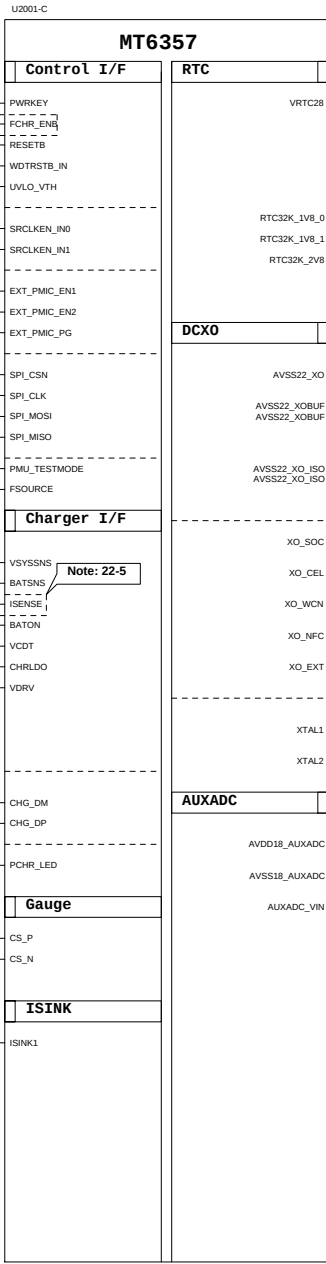
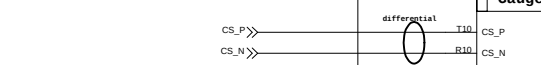
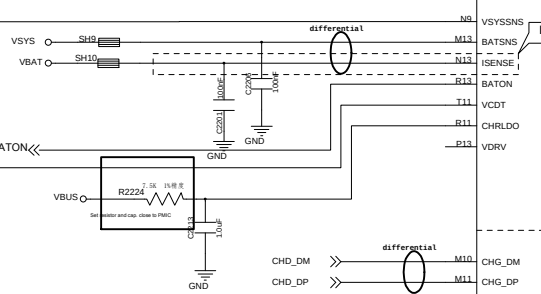
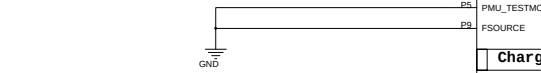
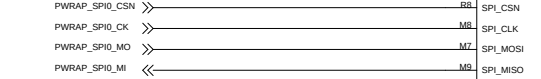
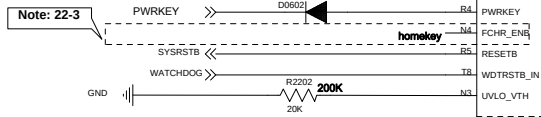


	Ext. buck option	
	w/ EXT VS2 Buck	w/o EXT VS2 Buck
C2104	10uF	22uF
R2851	0-ohm , 0603	NC
R2852	NC	0-ohm , 0603

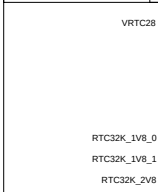
Note 21-5: Please connect VS2_LDO1(F15) to VS1_PMU if voltage applied to VCAMD(E17) ≥ 1.3 V

Schottky diode (VF <0.3V @ 1mA) MUST be for PWRKEY ESD protection and to avoid inverse current from MT6371.

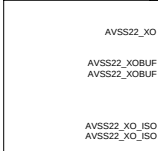
Note: 22-3



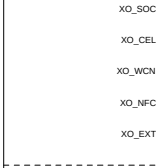
Control I/F



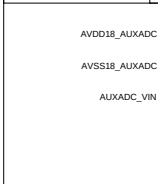
DCX0



Charger I/F



AUXADC



Gauge



ISINK



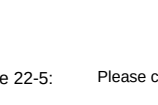
RTC



DCX0



Charger I/F



Gauge

ISINK

RTC

DCX0

Charger I/F

Gauge

ISINK

RTC

DCX0

Charger I/F

Gauge

ISINK

RTC

DCX0

Charger I/F

Gauge

ISINK

RTC

DCX0

Charger I/F

Gauge

ISINK

RTC

DCX0

Charger I/F

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DCX0

Charger I/F

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DCX0

Charger I/F

Gauge

ISINK

RTC

DCX0

Charger I/F

Gauge

ISINK

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Charger I/F

Gauge

ISINK

RTC

DCX0

Charger I/F

Gauge

ISINK

RTC

DCX0

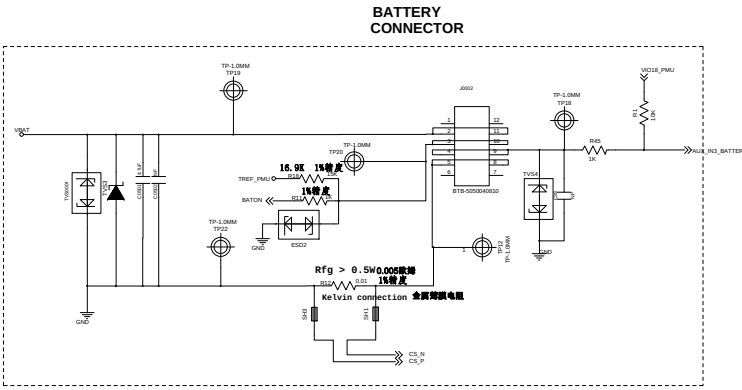
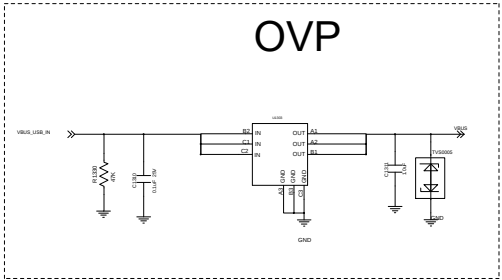
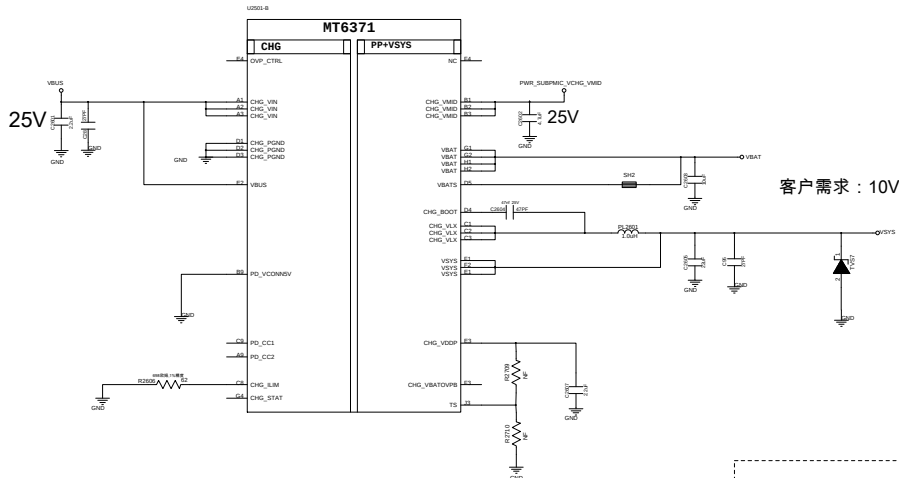
Charger I/F

Gauge

ISINK

RTC

DCX0

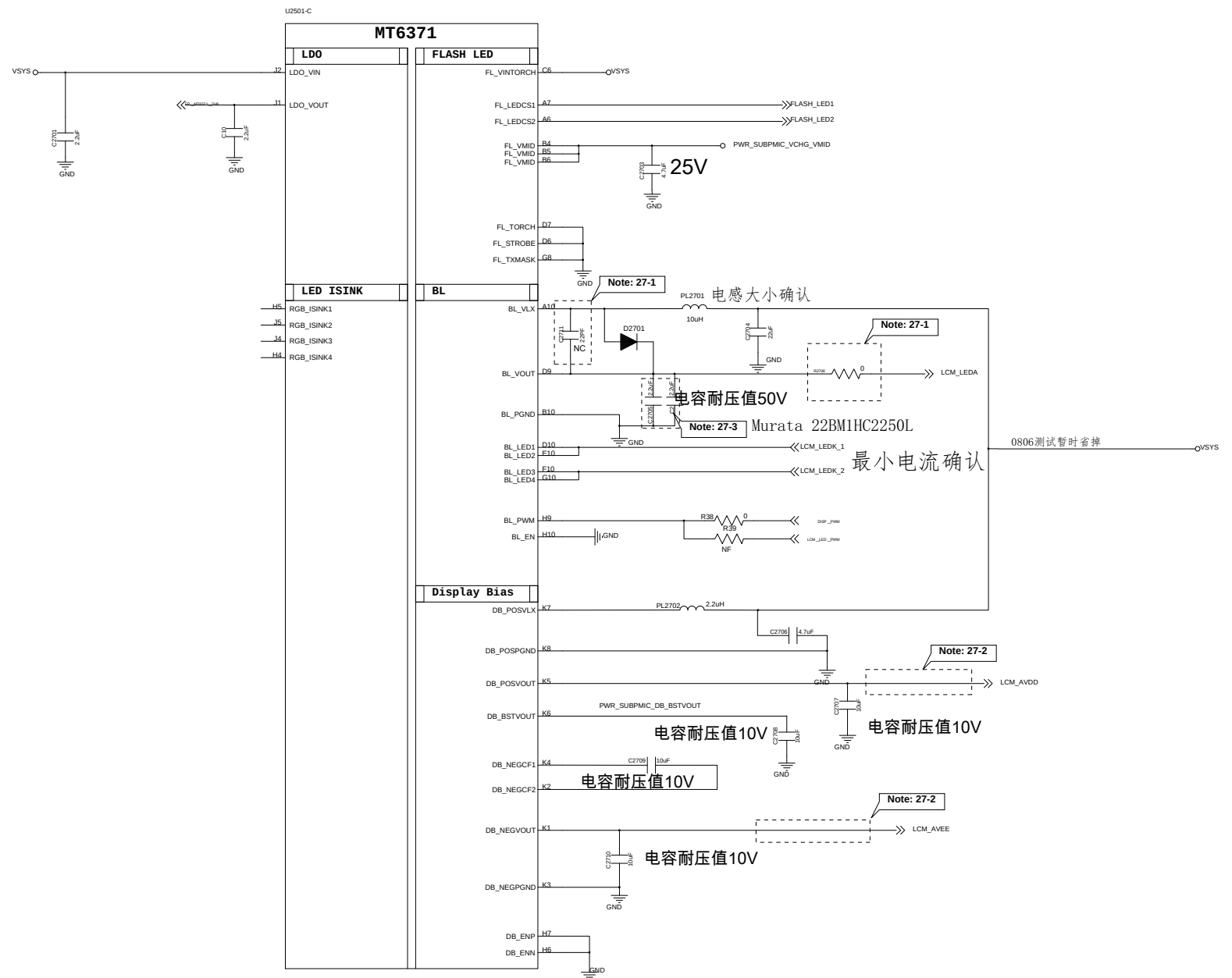


Note: 26-1

Schematic design notice of "26_POWER_MT6370-Charger + PP" page.

Note 26-1: For better ESD or surge performance we need choose suitable device for system protection. Please refer to [Surge device selection guide V2.0] provide by MTK.

Doc ID	26_POWER_SubPMIC-Charger + PP
Rev	MTK Confidential
Copyright	Copyright (c) 2014 MTK. All rights reserved.



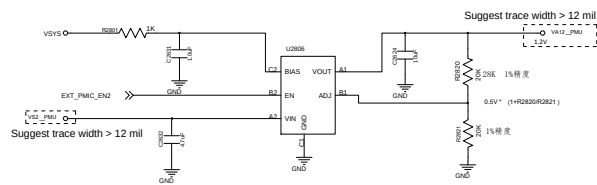
Schematic design notice of "27_POWER_SubPMIC-HV powers" page:

Note 27-1: To minimize RF de-sense, it is recommended to reserve 0-ohm and 0402 cap for BOM fine tuning.

Note 27-2: To minimize RF de-sense, it is recommended to reserve 0-ohm and 0201 cap. for BOM fine tuning.

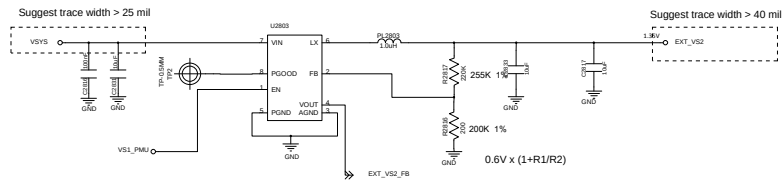
Note 27-3: C2705 could be replaced with C / 1 / uF / 50V + C / 1 / uF / 50V

LDO for VA12



Ext. Bulk for VS2

Note: 28-2



LDO for VCN33

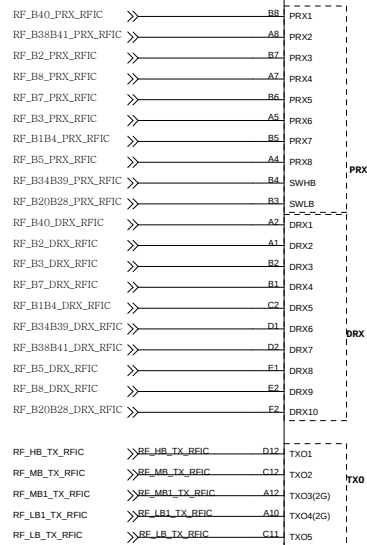
Please check MT67xx QVL

Schematic design notice of "28_POWER_ThirdParty-Power"

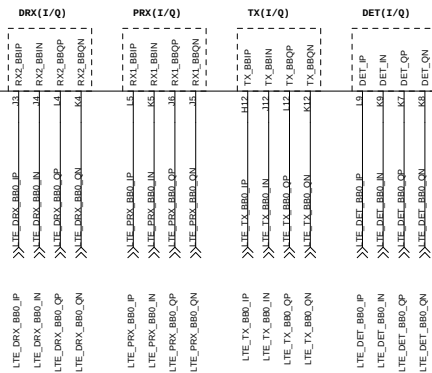
Note 28-1: VA12 Layout placement please close to AP

Note 28-2: VS2 Buck Layout placement please close to PMIC MT6357

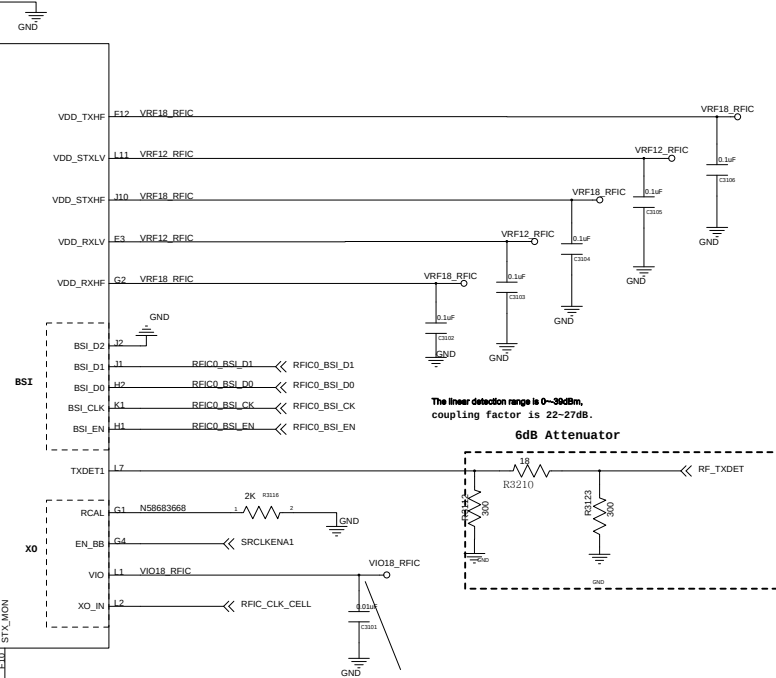
Note 28-3: VCN33 LDO Layout placement please close to MT6631



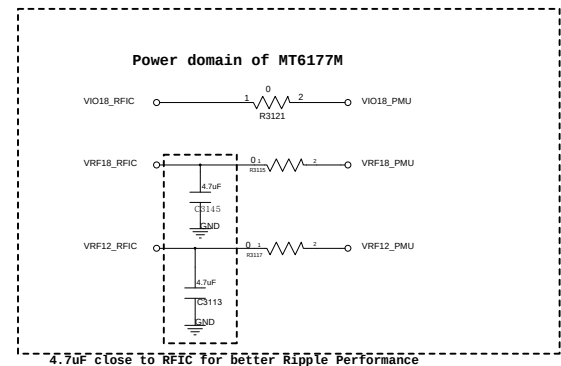
MT6177M

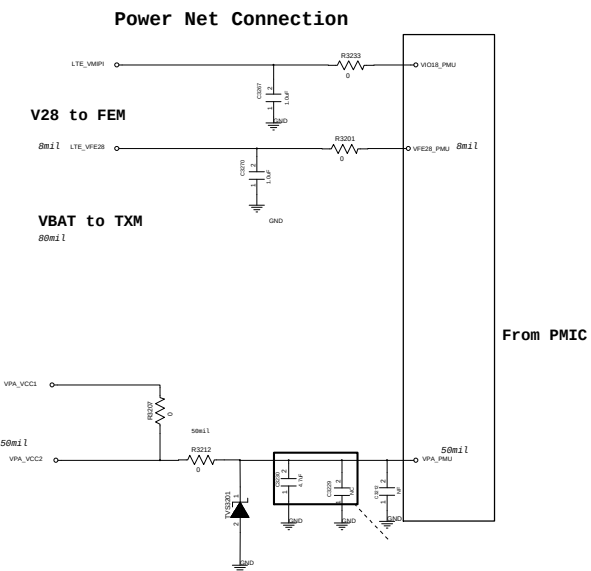


No Connection, for PN test



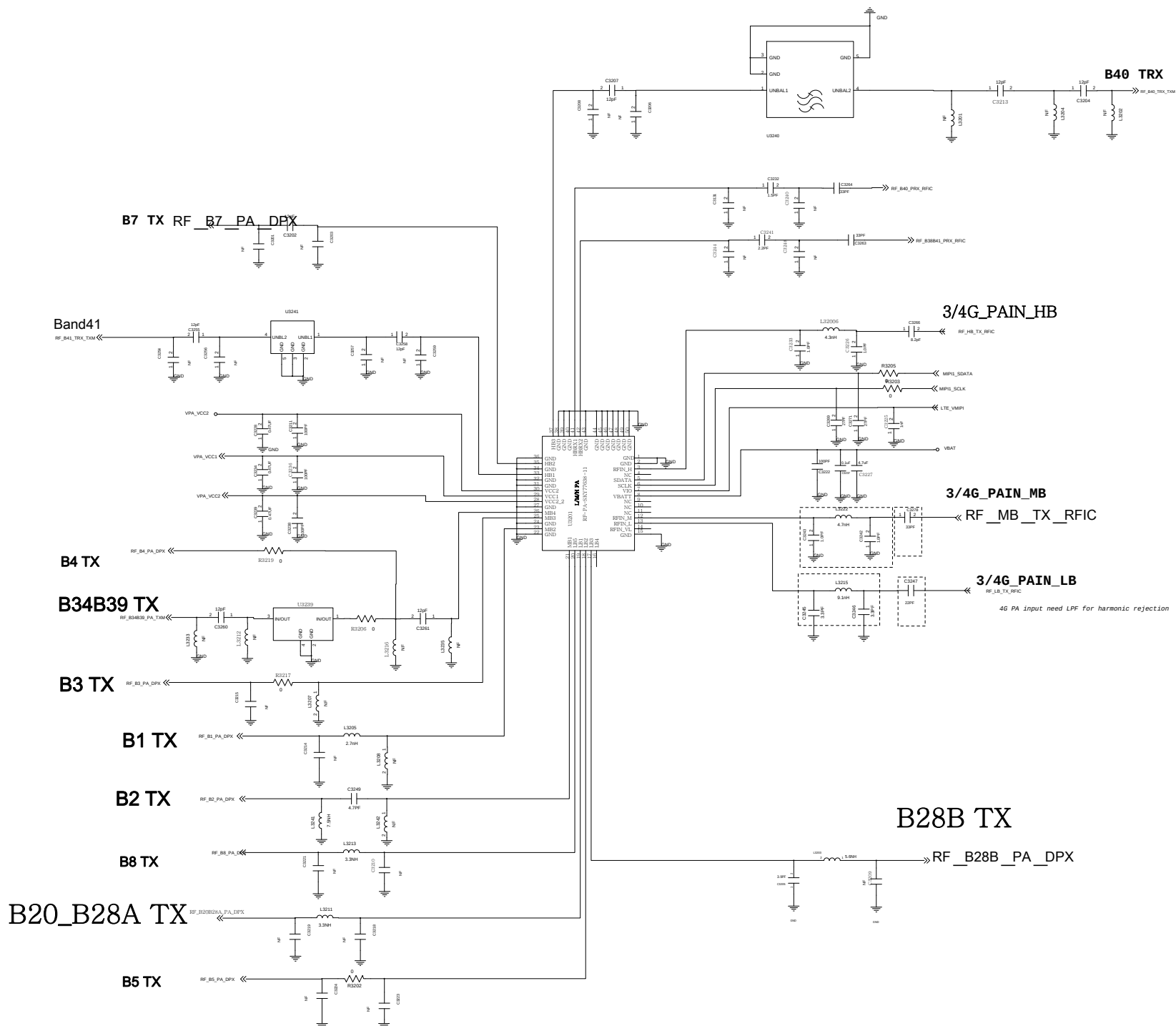
For phone, remove 0R to avoid low slew rate.
 For EVB, put test point.

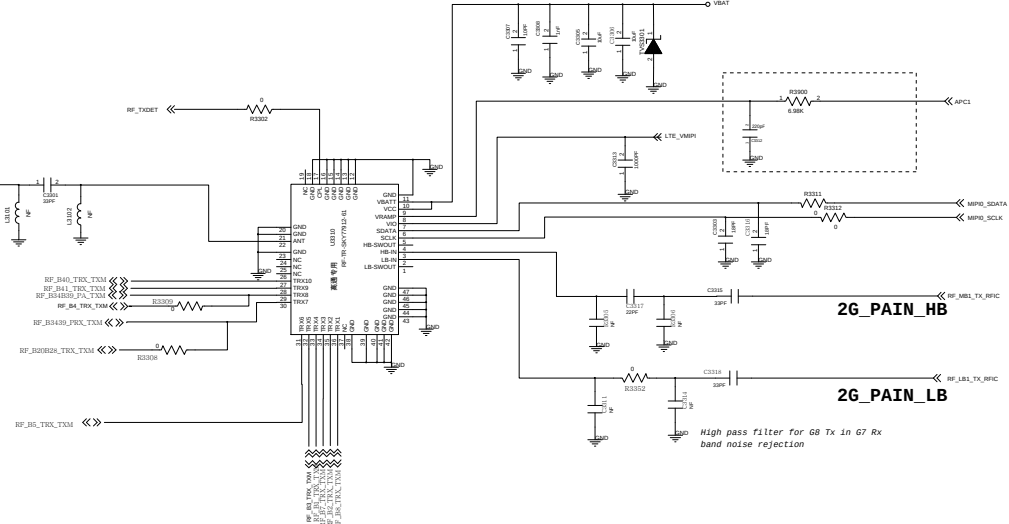




Note:
Use 0603 for lower Capacitance drop

Note:
Reserve at least one tuning cap. For PMIC
VPA total cap requirement, the total cap at
RF side should be in 6.2uF +/-5%.



[illegible][illegible]

B39 PRX

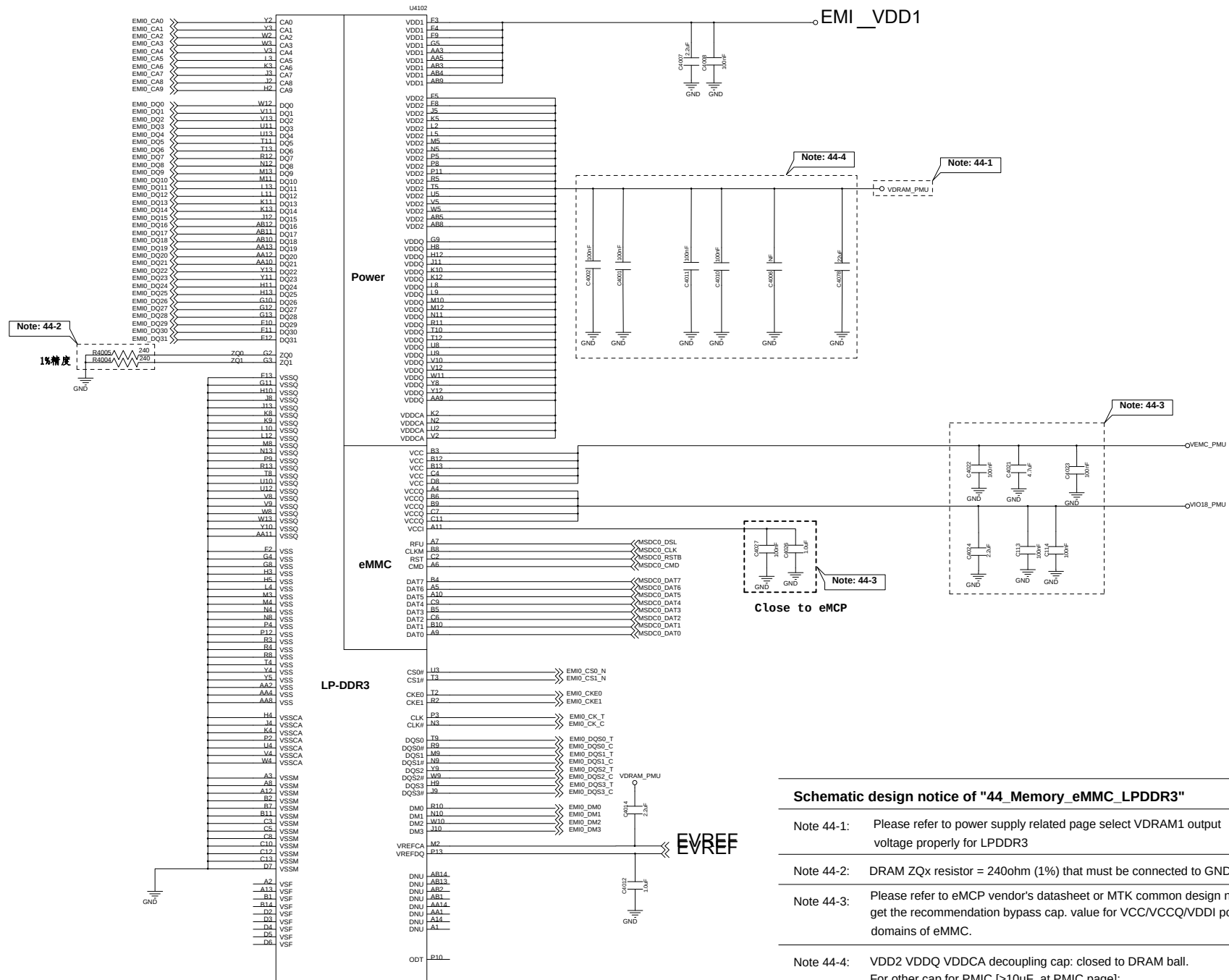
Band Pass Filter for SAWless Application.

```

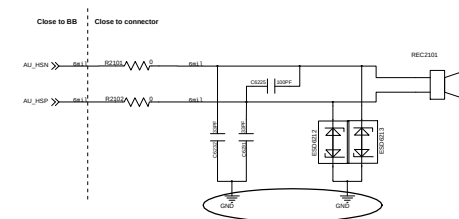
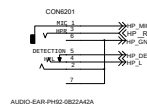
graph LR
    IN[IN_B39PRX_PRX_IN] -- 50 Ohm --> C1[100 pF]
    C1 --> C2[100 pF]
    C2 --> L3230[L3230]
    L3230 --> C3[100 pF]
    C3 --> C4[100 pF]
    C4 -- 50 Ohm --> C5[100 pF]
    C5 --> OUT[OUT_B39PRX_PRX_OUT]
  
```

[illegible]

eMMC+LPDDR3



Earphone EINT



Connecting the GNDs together and then to main GND through single via

选择ACC MODE

Note: 62-4

Platform	W6357		
Mode	ACC mode	DCC mode	
EINT	HP_EINT	HP_EINT	
Key-Detection	ACCDDET	AU_VIN1_P	
R6229	1uF	0 ohm	
R6230	1uF	0 ohm	
R6209	1K	NC	
R6210	1.5K	NC	
R6219	NC	2.49K	
R6231	4.7uF	0 ohm	
R6225	0 ohm	NC	
R6208	47K ohm	47K ohm	

Schematic design notice of "62_PERI_AUDIO_IO" page.

Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

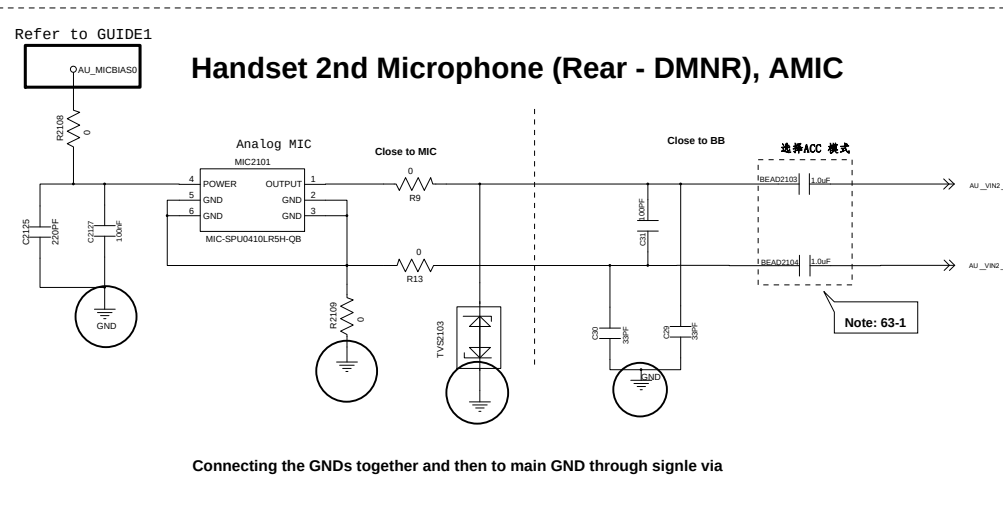
Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Note 62-3:

	Earphone Jack @ Main board	Earphone Jack @ Sub board
R0213	0 ohm	100nF

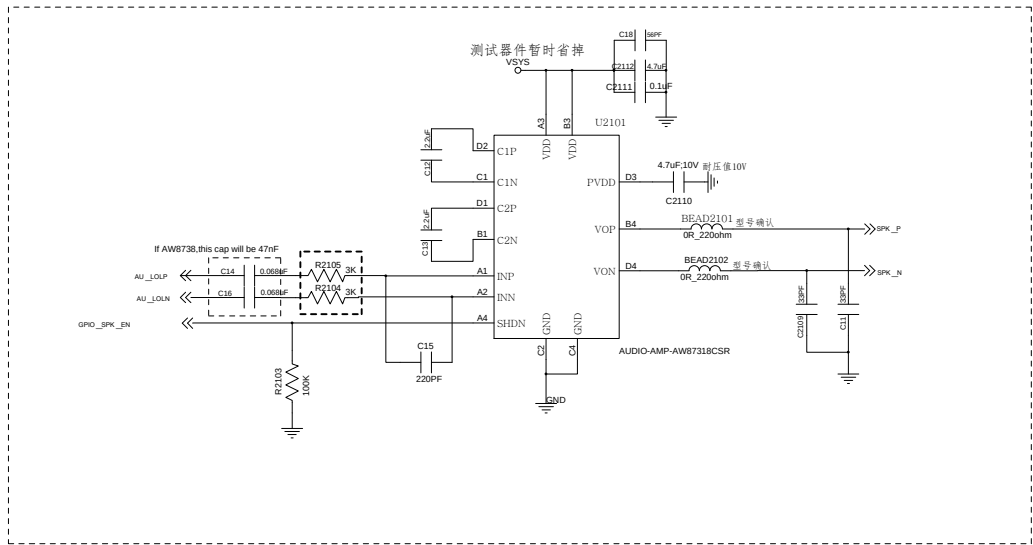
Note 62-4: Please Select ACC Mode for Operator Project to Pass Electrical MOS Test;
More Information refer to Audio/Speech Design Notice

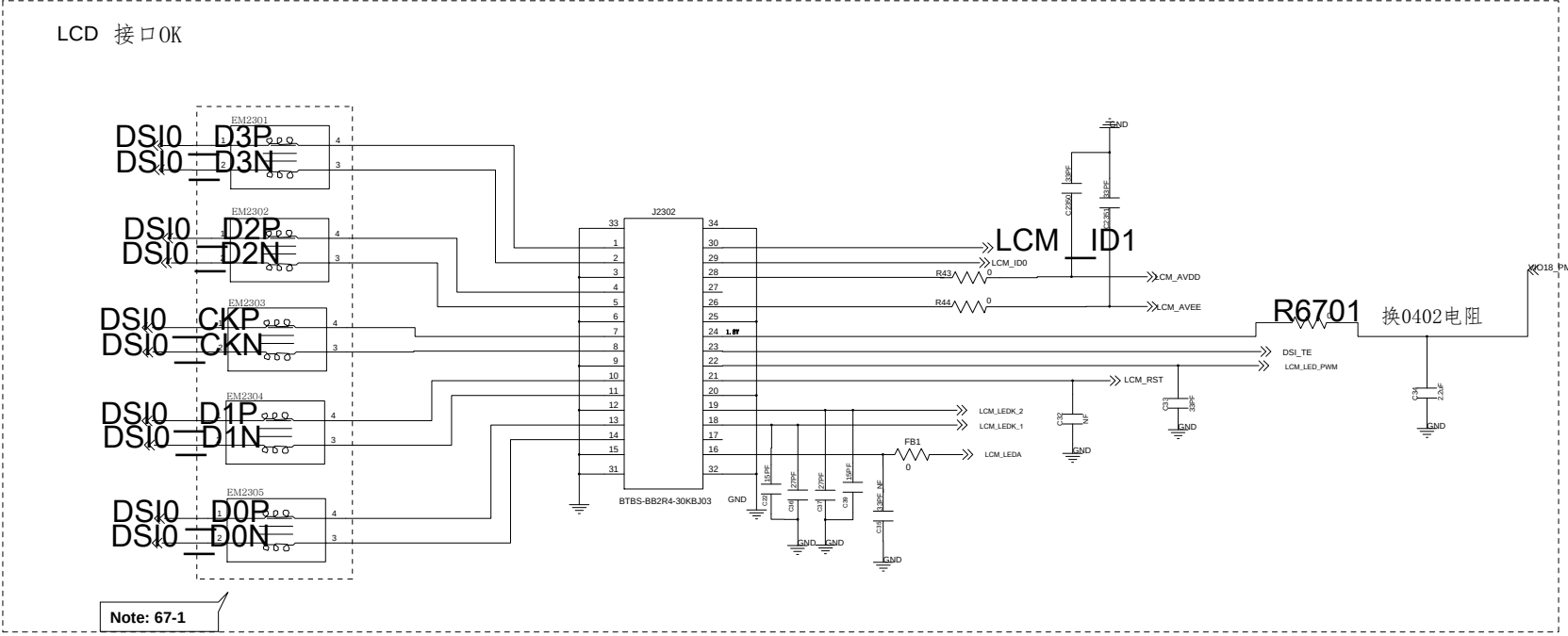
Note 62-5: Please select R6231 with 0402 size



Schematic design notice of "63_PERI_AUDIO_IO" page.

Note 63-1: 1 uF for ACC mode
0 ohm for DCC mode





Note: If best EMI practices are followed for MIPI CSI/DSI signals, there is no need for common mode choke filters. You may choose to have placeholders for common mode depending upon your design constraints. Extreme care must be taken that no stubs are created by doing so.

LCM ID

ID1 vender	EBBG
0	0
1	0
0	0

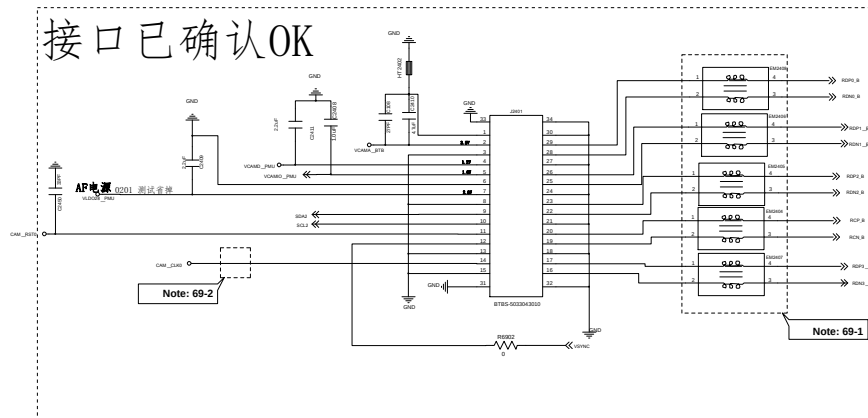
Schematic design notice of "67_PERI_LCD_CTP" page.

Note 67-1: It is recommended to reserve common-mode choke to prevent RF de-sense, the max. cap loading of common-mode choke must be less than 3pF.

供应商修改1 pin位置，需要按网络名核对

Main Camera

接口已确认OK



Schematic design notice of "69_PERI_CAMERA" page.

Note 69-1:

It is recommended to reserve common-mode choke to prevent RF de-sense,
the max. cap loading of common-mode choke must be less than 3pF.

Note 69-2:

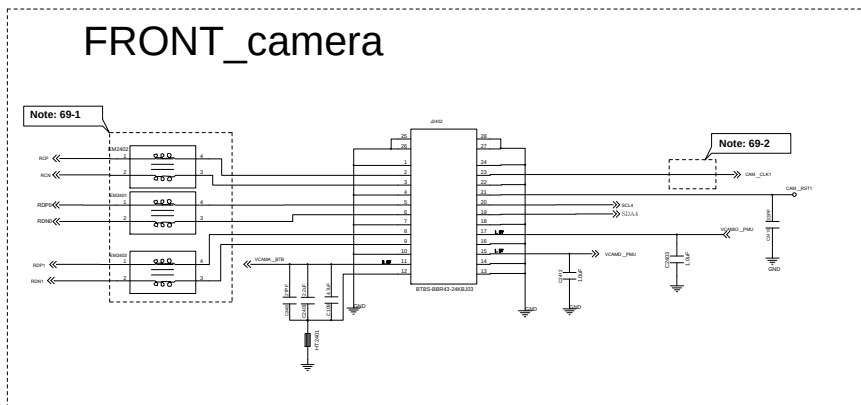
It is recommended to reserve 0-ohm for BOM fine tune to minimize RF de-sense.

测试器件暂时省掉



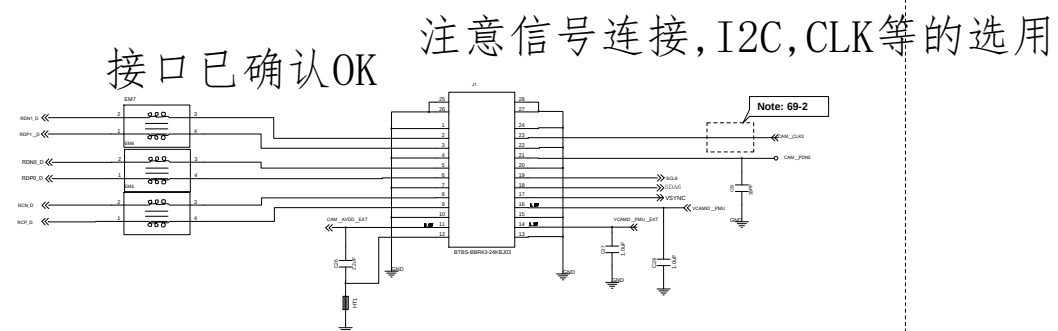
接口已确认OK

FRONT_camera

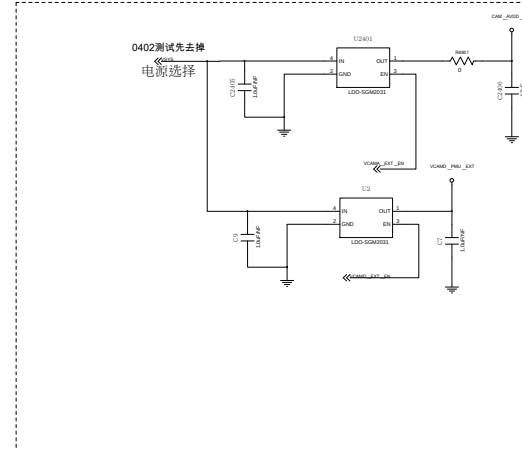


增加双摄

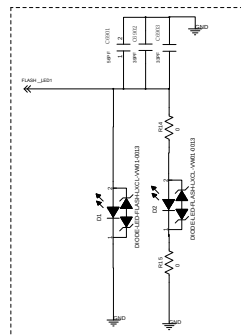
接口已确认OK



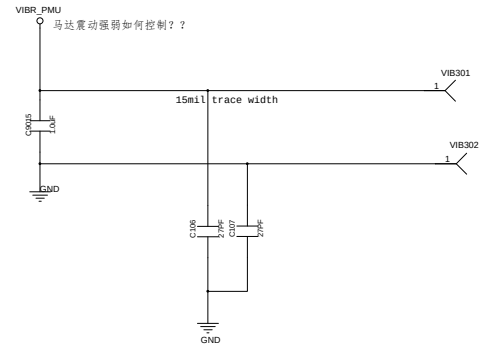
AVDD, DVDD需要单独增加LDO

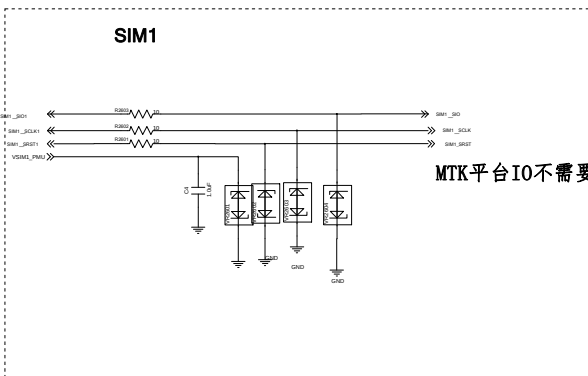


闪光灯

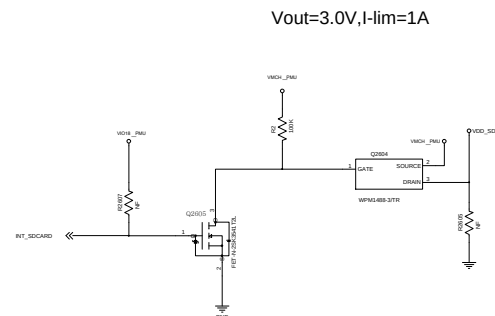


VIBRATOR



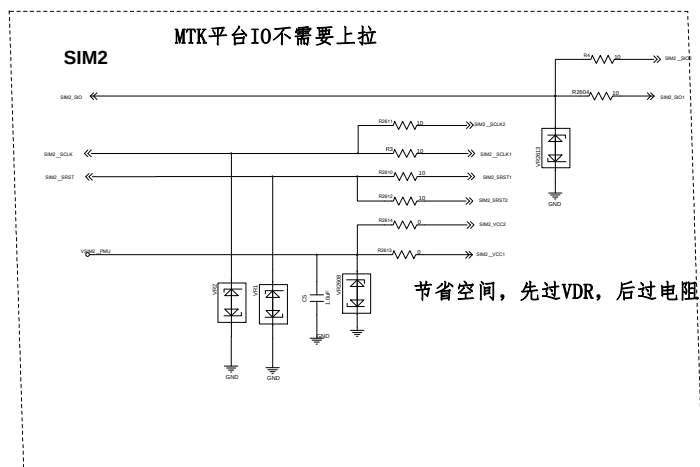
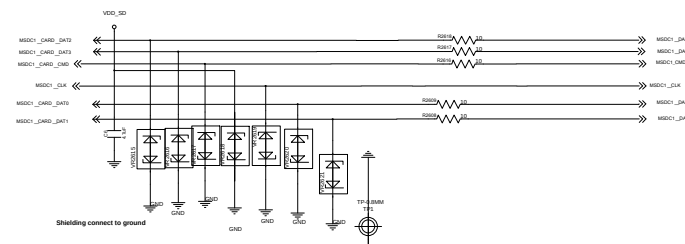


MTK平台IO不需要上拉



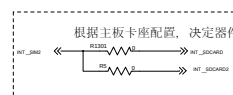
Vout=3.0V, I-lim=1A

SD CARD



MTK平台IO不需要上拉

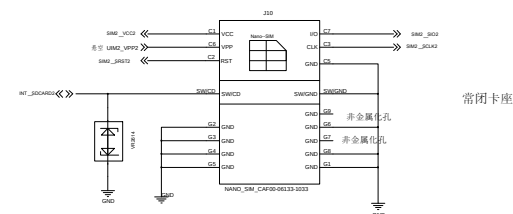
节省空间, 先过VDR, 后过电阻



NOTE:

For internal version, ONLY use J2601:
R2603/R2610/R2613/R2604 can be 0Ω, R2611/R2612/R2614/R2615 can be NF,
R1301 can be 0Ω, Q2601 NF; R1302 NF

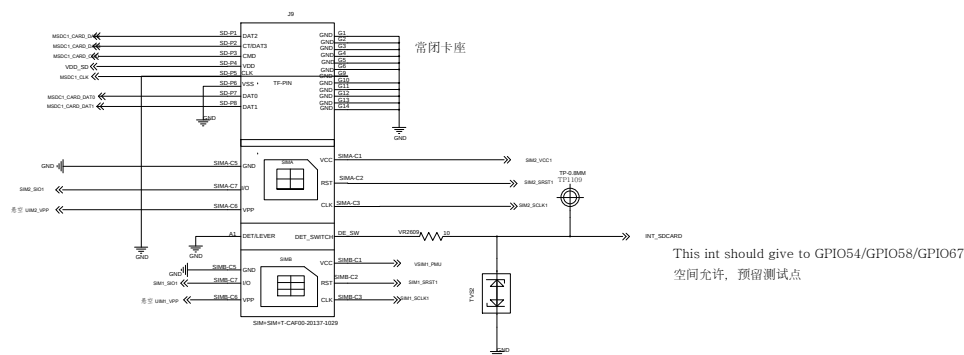
For India version:
R2611/R2612/R2614/R2615 can be 0Ω, R2603/R2610/R2613/R2604 can be NF,
Q2601 mount; R1301 NF; R1302 100K



常闭卡座

For internal version, R2603/R2610/R2613/R2604 can be 0Ω, R2611/R2612/R2614/R2615 can be NF,
For India version, R2611/R2612/R2614/R2615 can be 0Ω, R2603/R2610/R2613/R2604 can be NF.

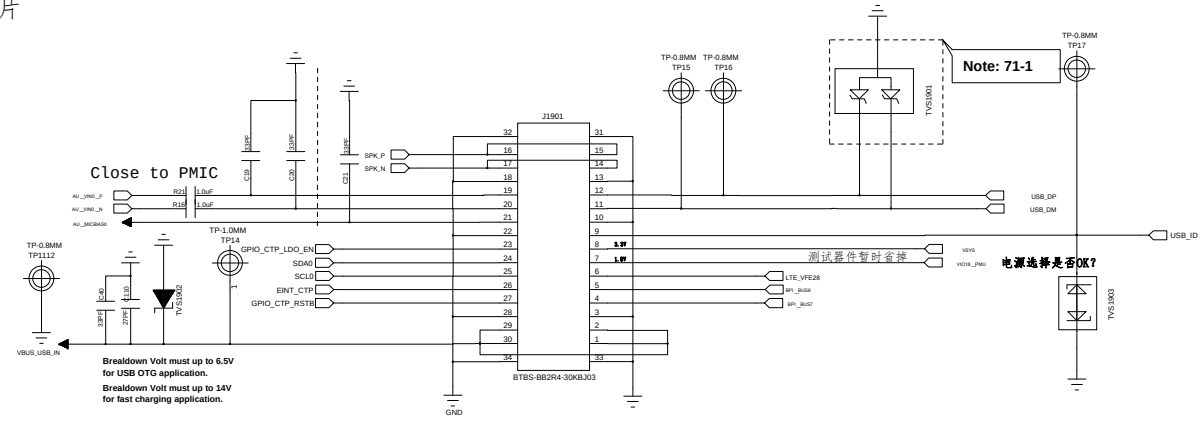
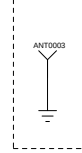
SIM/TF card



This int should give to GPIO54/GPIO58/GPIO67
空间允许, 预留测试点

M-MIC+USB+RGB+SPK

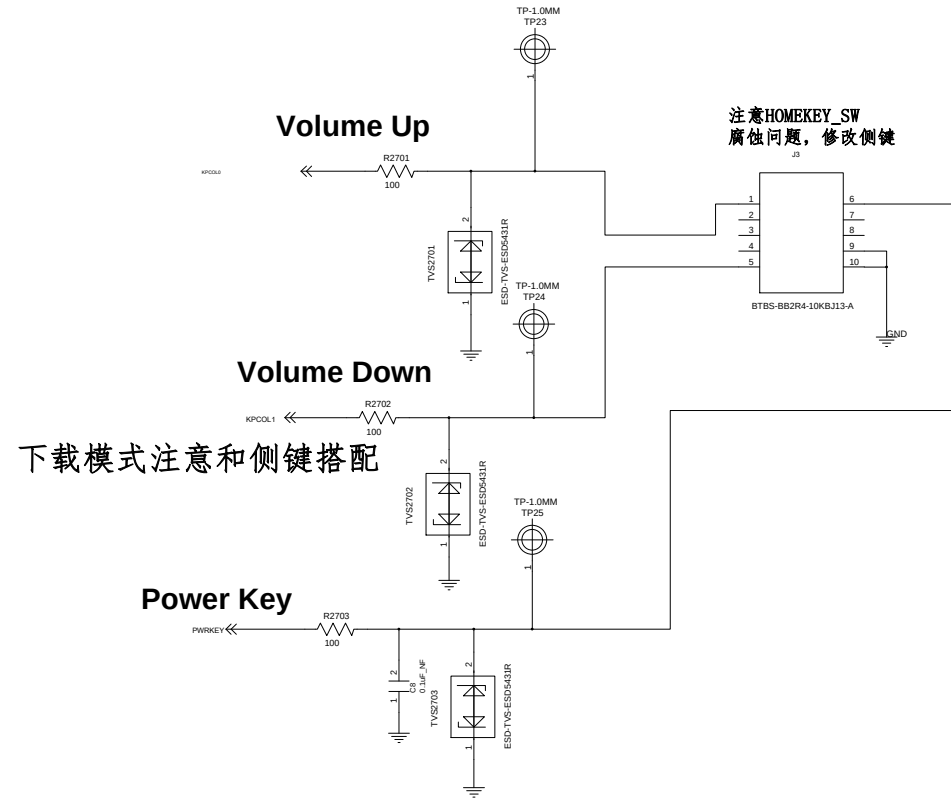
主板接地弹片



CTP接口

Power Key / Key Pad

DO NOT put pull-up resistor on PWRKEY



Schematic design notice of "65_PERI_Dual_SIM_ICUSB_KEYPAD" page.

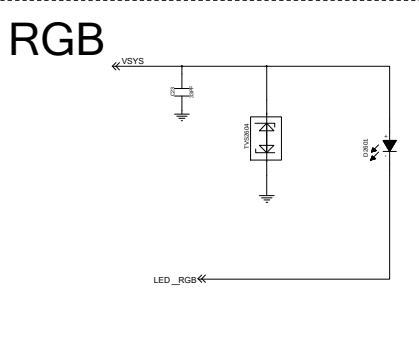
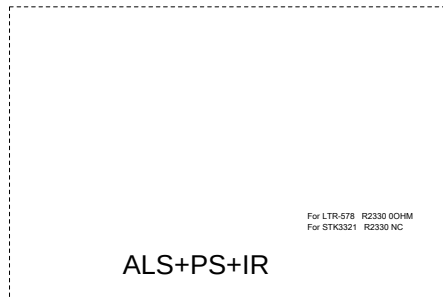
Note 75-1: DO NOT put pull-up resistor on PWRKEY

Note 75-2:

Volume Up : HOME Key / GND

Volume Down : KPROW0/KPCOL0

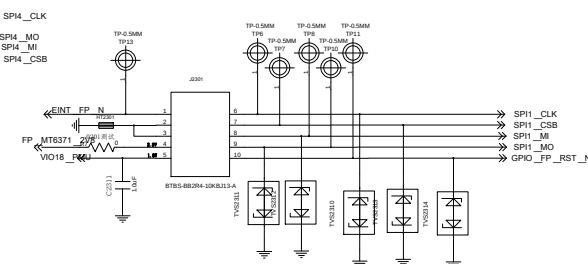
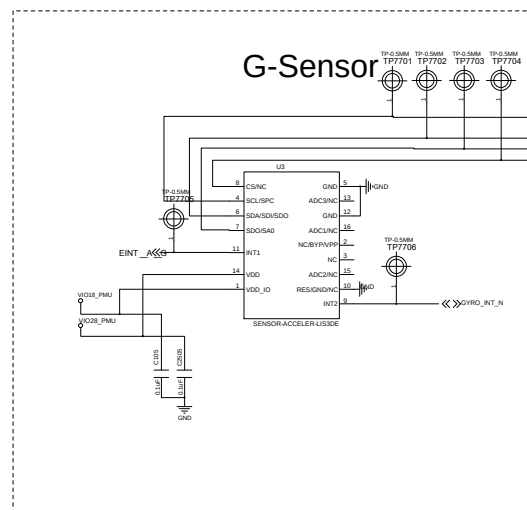
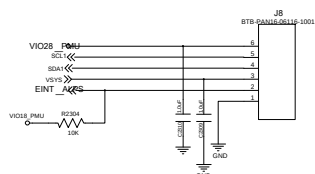
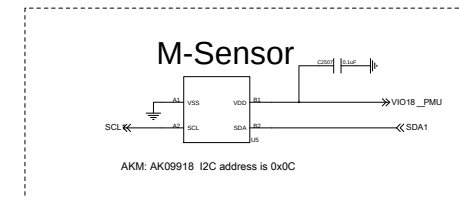
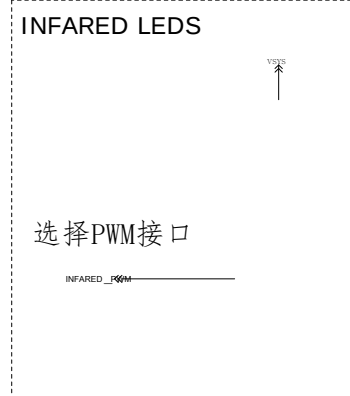
CM36558 INT default Output High, Active Low



Accerometer + Gyro Sensor

default SPI (if sensor hub support)

6DSx INT1/2 default Output Low, Active high



Replaced by fingerprint Sensor connector. follow 8996 placement

测试器件暂时省掉

Schematic design notice of "77_PERI_SENSORS_MEMS_ALS/PS" page.

Note 77-1: [M sensor] Keep a minimum distance of 15mm from power ICs / PCB traces of more than 100mA / magnet component. Check HW design notice for more detail.

Note 77-2: [A+G] For optimized GPS performance, please check HW design notice for Sensor selection guide

Note 77-3: [A+G] MUST use SPI for optimized sensor hub performance

DO NOT USE i2C

Note 77-4: [A+G] Suggest choose sensor support FIFO watermark interrupt otherwise we cannot support Hifi-sensor, daydream VR. And Sensor-location accuracy will become worse.

Note 77-5: [Baro] Reserve Baro sensor for LPPe feature (Must for North America Operator / NA SKU)

Note 77-6: DO NOT share Sensor hub i2C to other non-SCP device



FIDUCIAL-MARK-1MM FIDUCIAL-MARK-1MM FIDUCIAL-MARK-1MM FIDUCIAL-MARK-1MM

TP131 TP132 TP133 TP134



	屏蔽盖	屏蔽框
BB		
RF		
SUB-PMU		
WCN		